

CONTROL CODES · \$00-\$1F

| HEX | DEC | DESCRIPTION       | STD ASCII | # BYTES | NOTES                                                                                                 |
|-----|-----|-------------------|-----------|---------|-------------------------------------------------------------------------------------------------------|
| 00  | 0   | NULL              | Yes       | 1       | No operation                                                                                          |
| 01  | 1   | CURSOR HOME       | Yes       | 1       | Move cursor to 0,0                                                                                    |
| 02  | 2   | CURSOR CHARACTER  |           | 2       | 2nd byte is the cursor character, or \$00 to turn off <span>DEFAULT OFF</span>                        |
| 03  | 3   | CURSOR MODE       |           | 1       | Toggle solid / blinking <span>DEFAULT SOLID</span>                                                    |
| 04  | 4   | RESET             |           | 1       | Text mode, clear screen, cursor home / off / solid, BG \$00, FG \$FF                                  |
| 05  | 5   | BELL DURATION     |           | 2       | 2nd byte is duration in jiffies (1/60 s) <span>DEFAULT \$3C</span>                                    |
| 06  | 6   | BELL FREQUENCY    |           | 2       | 2nd byte is a note index — see Bell Frequencies <span>DEFAULT \$3D</span>                             |
| 07  | 7   | BELL              | Yes       | 1       | Queue one bell at the current duration and frequency                                                  |
| 08  | 8   | BACKSPACE         | Yes       | 1       | Cursor left one column; stops at column 0                                                             |
| 09  | 9   | TAB               | Yes       | 1       | Advance to the next 4-column stop                                                                     |
| 0A  | 10  | LINE FEED         | Yes       | 1       | Cursor down, same column; scrolls up at the last row                                                  |
| 0B  | 11  | SCREEN MODE       |           | 1       | Toggle text / graphics <span>DEFAULT TEXT</span>                                                      |
| 0C  | 12  | CLEAR SCREEN      | Yes       | 1       | Fill the screen with the background color                                                             |
| 0D  | 13  | CARRIAGE RETURN   | Yes       | 1       | Cursor to column 0, same row                                                                          |
| 0E  | 14  | SET COLUMN        |           | 2       | 2nd byte is the column, taken modulo the column count — 0–39 at 40, 0–79 at 80 <span>DEFAULT 0</span> |
| 0F  | 15  | SET ROW           |           | 2       | 2nd byte is the row, taken modulo the row count — 0–29 at 40, 0–59 at 80 <span>DEFAULT 0</span>       |
| 10  | 16  | DELETE TO SOL     |           | 1       | Clear from the start of the line through the cursor                                                   |
| 11  | 17  | DELETE TO EOL     |           | 1       | Clear from the cursor to the end of the line                                                          |
| 12  | 18  | DELETE TO SOS     |           | 1       | Clear from the start of the screen through the cursor                                                 |
| 13  | 19  | DELETE TO EOS     |           | 1       | Clear from the cursor to the end of the screen                                                        |
| 14  | 20  | SCROLL LEFT       |           | 1       | Vacated column filled with the background color                                                       |
| 15  | 21  | SCROLL RIGHT      |           | 1       | Vacated column filled with the background color                                                       |
| 16  | 22  | SCROLL UP         |           | 1       | Vacated row filled with the background color                                                          |
| 17  | 23  | SCROLL DOWN       |           | 1       | Vacated row filled with the background color                                                          |
| 18  | 24  | FOREGROUND COLOR  |           | 2       | 2nd byte is RGB332 color data <span>DEFAULT \$FF WHITE</span>                                         |
| 19  | 25  | BACKGROUND COLOR  |           | 2       | 2nd byte is RGB332 color data <span>DEFAULT \$00 BLACK</span>                                         |
| 1A  | 26  | NEXT BYTE AS DATA |           | 2       | Lets \$00–\$1F and \$7F be interpreted as data rather than command                                    |
| 1B  | 27  | ESC               | Yes       | 2       | Introduces a two-byte extension — see ESC Extensions                                                  |
| 1C  | 28  | CURSOR LEFT       |           | 1       | Stops at column 0                                                                                     |
| 1D  | 29  | CURSOR RIGHT      |           | 1       | Stops at the last column                                                                              |
| 1E  | 30  | CURSOR UP         |           | 1       | Stops at row 0                                                                                        |
| 1F  | 31  | CURSOR DOWN       |           | 1       | Stops at the last row                                                                                 |

CHARACTER DATA · \$20-\$FF

| HEX   | DEC     | DESCRIPTION         | STD ASCII | # BYTES | NOTES                                 |
|-------|---------|---------------------|-----------|---------|---------------------------------------|
| 20–7E | 32–126  | ASCII CHARACTERS    | Yes       | 1       | See ASCII chart                       |
| 7F    | 127     | DELETE              | Yes       | 1       | Clear the cell at the cursor          |
| 80–FF | 128–255 | EXTENDED CHARACTERS |           | 1       | CP437 upper range — see character set |

ESC EXTENSIONS · \$1B + 1 BYTE

v1.3.0 reserved \$1B for future escape codes and treated it as a no-op. It is now the introducer for two-byte extensions. An unrecognised second byte is processed as an ordinary byte, so a stream that carried a stray ESC behaves exactly as it did under v1 — the deviation is the five sequences below and nothing else.

| SEQUENCE | DESCRIPTION        | # BYTES | NOTES                                                                                                          |
|----------|--------------------|---------|----------------------------------------------------------------------------------------------------------------|
| 1B 01    | 40-COLUMN MODE     | 2       | 320 × 240 pixels, 40 × 30 cells. Clears the screen and homes the cursor <input type="button" value="DEFAULT"/> |
| 1B 02    | 80-COLUMN MODE     | 2       | 640 × 480 pixels, 80 × 60 cells. Clears the screen and homes the cursor                                        |
| 1B 03    | VT-100 PERSONALITY | 2       | Hand the stream to the ANSI / VT-100 parser                                                                    |
| 1B 04    | QUERY              | 2       | Reply on the serial link with personality, columns and rows                                                    |
| 1B 1B    | LITERAL ESC        | 2       | Emit \$1B as data                                                                                              |
| 1B *     | NO-OP ESC          | 2       | Any other second byte runs as if the ESC were not there — v1's behavior                                        |

The query reply is five bytes — 1B 04, then personality, then the column and row counts as literal numbers: 1B 04 00 28 1E for 40 × 30, 1B 04 00 50 3C for 80 × 60. Personality always reads 00 (native): the query is itself a native extension, so a terminal in VT-100 mode never reaches it.

SCREEN GEOMETRY

| MODE      | PIXELS    | CELLS   | CELL  |
|-----------|-----------|---------|-------|
| 40-COLUMN | 320 × 240 | 40 × 30 | 8 × 8 |
| 80-COLUMN | 640 × 480 | 80 × 60 | 8 × 8 |

An exact 2× of the 40-column grid in both axes — same 8 × 8 font, square pixels, 4:3 aspect, and 8 rows per cell in graphics mode.

PERSONALITIES

| MODE   | PROTOCOL                                         | ENTER       |
|--------|--------------------------------------------------|-------------|
| NATIVE | This card <input type="button" value="DEFAULT"/> | ESC [?7000h |
| VT-100 | ANSI / VT-100                                    | 1B 03       |

Native is byte-for-byte v1.3.0 plus the ESC extensions above. VT-100 mode is documented separately — see the README. Each personality is left by a sequence the *other* one cannot express, so neither is a one-way door. ESC c (RIS) returns to whichever was configured at launch.

BELL FREQUENCIES · 2ND BYTE OF \$06

Note index \$01–\$54, C1 through B7. Any value outside that range silences the bell, as does a duration of \$00. Default is \$3D — C6, 1046.50 Hz.

| OCT | C  | C# | D  | D# | E  | F  | F# | G  | G# | A  | A# | B  |
|-----|----|----|----|----|----|----|----|----|----|----|----|----|
| 1   | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 0A | 0B | 0C |
| 2   | 0D | 0E | 0F | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 3   | 19 | 1A | 1B | 1C | 1D | 1E | 1F | 20 | 21 | 22 | 23 | 24 |
| 4   | 25 | 26 | 27 | 28 | 29 | 2A | 2B | 2C | 2D | 2E | 2F | 30 |
| 5   | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 3A | 3B | 3C |
| 6   | 3D | 3E | 3F | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| 7   | 49 | 4A | 4B | 4C | 4D | 4E | 4F | 50 | 51 | 52 | 53 | 54 |

Highlighted: \$2E A4 = 440.00 Hz · \$3D C6 = 1046.50 Hz (default). Duration is in jiffies — \$3C = 60 = one second.